

Inheritance is one in which a derived class inherits the already existing class's features.

Polymorphism is one that you can define in different forms.

It refers to using the structure and behavior of a superclass in a subclass.

It refers to changing the behavior of a superclass in the subclass.

It is required in order to achieve polymorphism.

In order to achieve polymorphism, inheritance is not required.

It is applied to classes.

It is applied to functions and methods.

It can be single, hybrid, multiple, hierarchical, multipath, and multilevel inheritance.

There are two types of polymorphism compile time and run time.

It supports code reusability and reduces lines of code.

It allows the object to decide which form of the function to be invoked at run-time (overriding) and compile-time (overloading).

28) What is Coupling in OOP and why it is helpful?

In programming, separation of concerns is known as coupling. It means that an object cannot directly change or modify the state or behavior of other objects. It defines how closely two objects are connected together. There are two types of coupling, loose coupling, and tight coupling.

Objects that are independent of one another and do not directly modify the state of other objects is called loosely coupled. Loose coupling makes the code more flexible, changeable, and easier to work with.